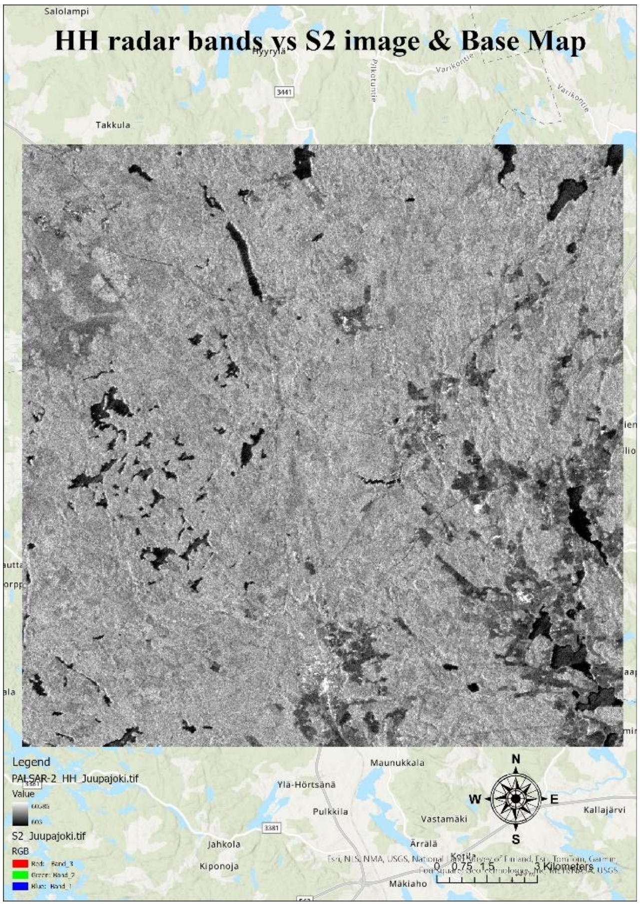
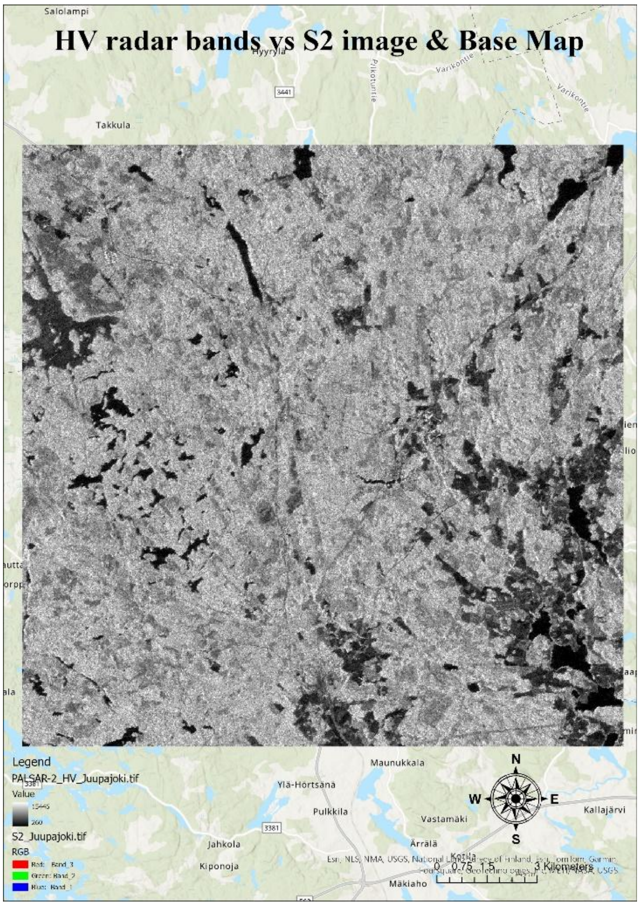
FORESTS INVENTORY WITH SATELLITE IMAGE DATA

TASK 1

Results of comparing the PALSAR-2 HH/HV bands and the land cover product against the base map as requested in the task.





Differences in backscattering intensity and patterns between the radar bands and the optical image is considered while comparing HH and HV polarized radar bands against the Sentinel-2 image.

Distinguishing targets from the radar images

In the above radar images, waterbodies, non-vegetation areas, and vegetation are the most easily distinguishable targets. Waterbodies, due to their low backscatter, appear as dark and non-vegetation areas, and due to their high backscatter (due to surface roughness), they appear as light in both HH and HV radar bands. Vegetation, on the other hand, shows different backscatter characteristics depending on vegetation type and structure.

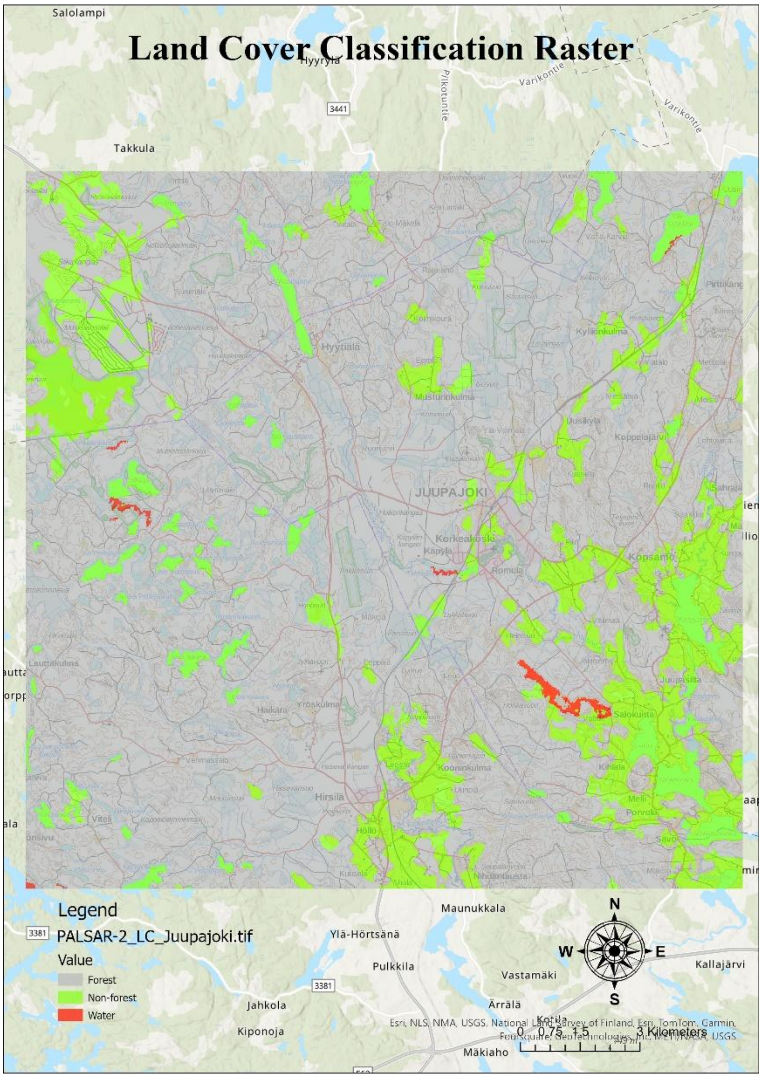
Recognition of various details of different forest types

Dense forests, due to their increased attenuation of radar signals within the canopy, exhibit relatively low backscattering compared to sparse forests and thus appear dark. Different forest types have varied vegetation density, height, and moisture content and thus can be recognized based on the backscatter intensity and texture in the radar images.

Practical application of the HH & HV radar bands in forest inventory

Both HH and HV polarized radar bands can be useful for forest inventory due to their sensitivity to forest structure and biomass. Various information on forest canopy density and structure can be obtained from HH polarization due to its sensitivity to the vertical structure of vegetation. HV polarization, on the other hand, can be sensitive to surface roughness and moisture content, which can be indicative of forest types and conditions.

Results of comparing the land cover product against the base map



The land cover classification raster includes classes for forests, which are represented by grey colour; non-forest, which is represented by green colour; and water, which is in red. The visual inspection alongside the base map reveals the fact that forested areas are accurately classified, and non-forested regions are also more or less accurately captured. However, forest characteristics such as vegetation density and structure are not as evident in comparison with the PALSAR bands.

Waterbodies are mostly misclassified as non-forest or forests or even vice versa since there are issues with spectral separability or the radar signal is getting attenuated differently in comparison to other land covers. Misclassified forest areas may also appear as non-forest or water if they have similar radar backscatter characteristics.

However, the erroneously classified targets can be recognised from the raw HH and HV radar bands due to the differences in backscatter intensity and texture between accurately classified and misclassified areas. As mentioned before, features such as strong backscatter in HH and HV bands may indicate non-forested areas, while low backscatter may suggest water bodies. The forested areas show different backscatter characteristics depending on vegetation type and structure.

TASK 2

The variable importance plot and its interpretation

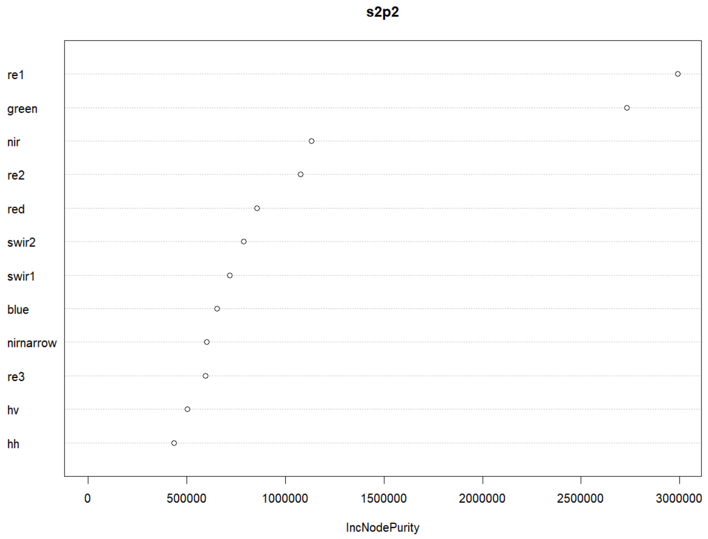


Figure 1: Variable importance plot of Sentinel-2 + PALSAR-2 model

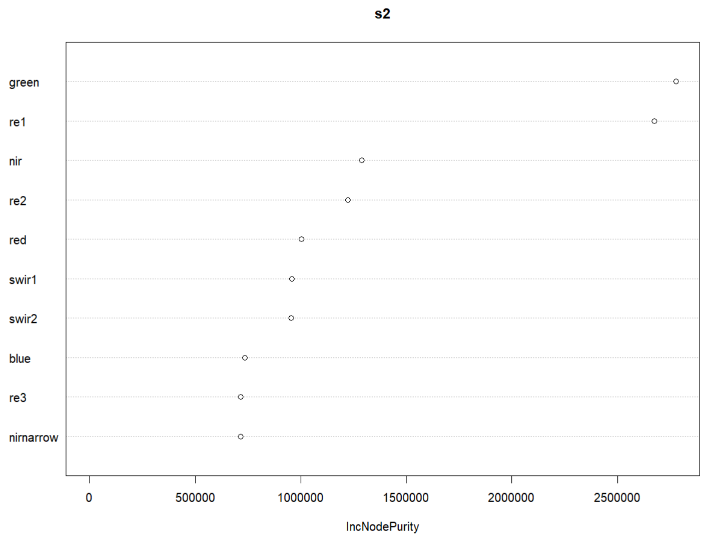


Figure 2: Variable importance plot of Sentinel-2 model

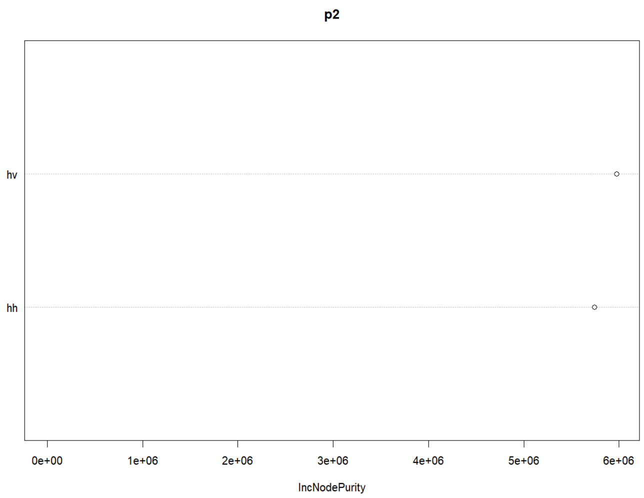


Figure 3: Variable importance plot of PALSAR-2 model

The importance of different predictors in model fitting using Random Forest can be inferred through variable importance plots. These plots indicate the relative significance of each variable in contributing to the accuracy of volume estimation, and the most important variables (those most frequently selected for node splits) have a larger score.

In the "S2P2" model (Figure 1), variables from both Sentinel-2 (blue, green, red, re1, re2, re3, nir, nirnarrow, swir1, swir2) and PALSAR-2 (HH and HV) data are included. From the variable importance plot, it can be observed that “re1” and “green” show higher importance scores compared to others.

In contrast, the variable importance plots for the "S2" and "P2" models (Figures 2 and 3, respectively) only show the importance of Sentinel-2 and PALSAR-2 variables, respectively. These plots provide insights into which type of data contributes more significantly to volume estimation when used individually. In the S2 model, “re1” and “green” show higher importance scores compared to others, just like in the S2P2 model, and in the P2 model, “hv” shows a relatively higher significance than “hh.”

RMSEs and scatter plots of predicted vs. observed values in the field data

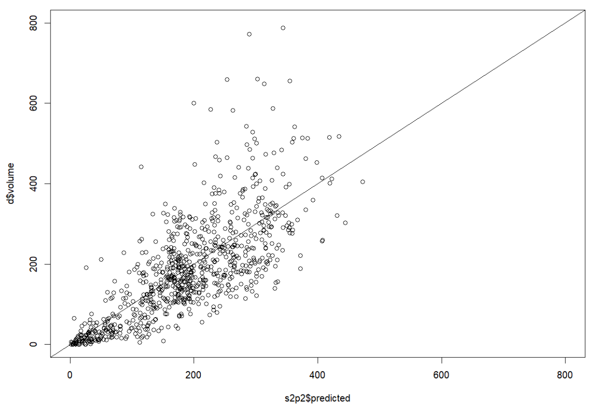


Figure 4: Scatter plot of predicted vs. observed values in Sentinel-2 + PALSAR-2 model

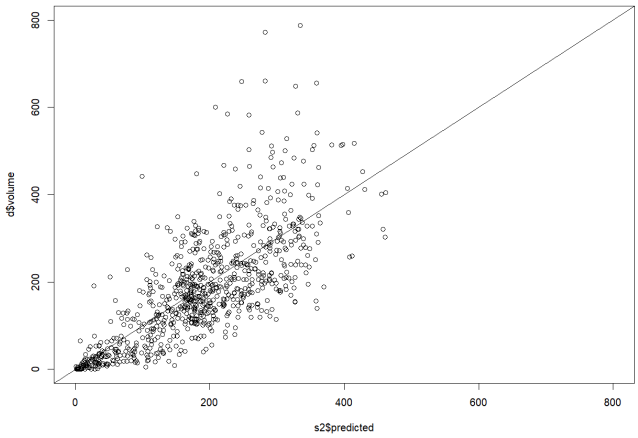


Figure 5: Scatter plot of predicted vs. observed values in Sentinel-2 model

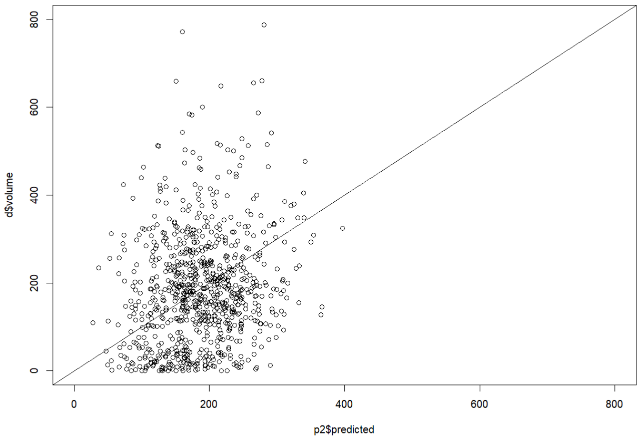


Figure 6: Scatter plot of predicted vs. observed values in PALSAR-2 model

The scatter plot depicted above displays the relationship between the predicted volume and the observed volume (on the y-axis) in three different models. The points are more or less closely aligned with the diagonal line, indicating the accuracy of the model prediction is quite good. The scatter plots do reveal some potential outliers, especially in the P2 model, that could affect the model's performance, and that could pave the way for further investigation.

Overall, most of the points align within and around the diagonal line and do not reveal any trend or pattern except for the P2 model. This shows that the models except P2 are quite accurate and suggests that the overall performance of the models is fine. In certain areas, it is evident that there is a slight spread in deviations from the diagonal line that shows areas where the model's performance is slightly inferior, and further investigation or model refinement can be done if we want the model to be a bit more accurate.

```
> rmse(d$volume, s2p2$predicted)
[1] 85.68368
> relrmse(d$volume, s2p2$predicted)
[1] 0.4596187
```

```
> rmse(d$volume, s2$predicted)
[1] 86.17206
> relrmse(d$volume, s2$predicted)
[1] 0.4622384
```

```
> rmse(d$volume, p2$predicted)
[1] 130.4512
> relrmse(d$volume, p2$predicted)
[1] 0.6997576
```

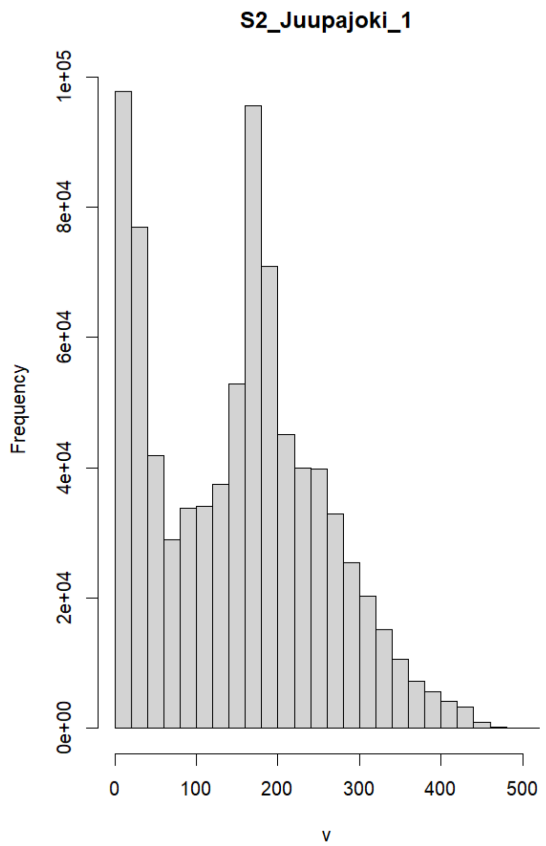
The results obtained for the RMSE and Relative RMSE provide insights into the performance and accuracy of the Random Forest model for predicting the volume. The "S2P2" model yielded an RMSE of 85.68 and a relative RMSE of 0.4596, indicating a reasonably accurate prediction of volume values. The "S2" model resulted in an RMSE of 86.17 and a relative RMSE of 0.4622, comparable to the "S2P2" model. However, the "P2" model exhibited a higher RMSE of 130.45 and a relative RMSE of 0.6998, indicating poorer performance in volume estimation when using only PALSAR-2 variables.

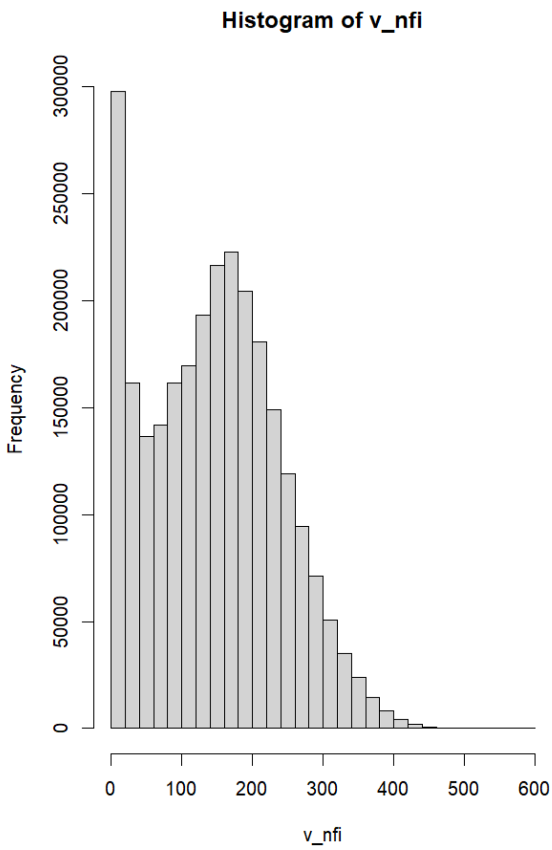
Evaluation and Conclusion:

From the variable importance plots, it appears that both Sentinel-2 and PALSAR-2 variables contribute to volume mapping in the Orivesi inventory area. However, when used individually, Sentinel-2 variables seem to perform slightly better than PALSAR-2 variables, as indicated by comparable RMSE values and relative RMSE percentages.

The "S2P2" model, which combines both Sentinel-2 and PALSAR-2 variables, provides the most accurate volume estimates, suggesting that a combination of optical and radar data enhances the accuracy of volume mapping. Therefore, for volume mapping in the Orivesi inventory area, utilising variables from both Sentinel-2 and PALSAR-2 datasets in a random forest model yields the most accurate results, with Sentinel-2 variables playing a crucial role in improving volume estimation accuracy.

Mean volumes and volume histograms (m3 /ha) from the s2p2 model and from the NFI map for the whole Juupajoki area





For s2p2 model, Mean volume = 152.1918 m³/ha

For NFI, Mean volume = 145.6553 m³/ha

Mean Volumes:

The mean volume predicted by the S2P2 model is 152.1918 m^3/ha. Based on the model's predictions, this value represents the average volume estimation across the Juupajoki area.

On the other hand, the mean volume derived from the NFI map is 145.6553 m^3/ha. This value reflects the average volume estimated by the National Forest Inventory for the same area.

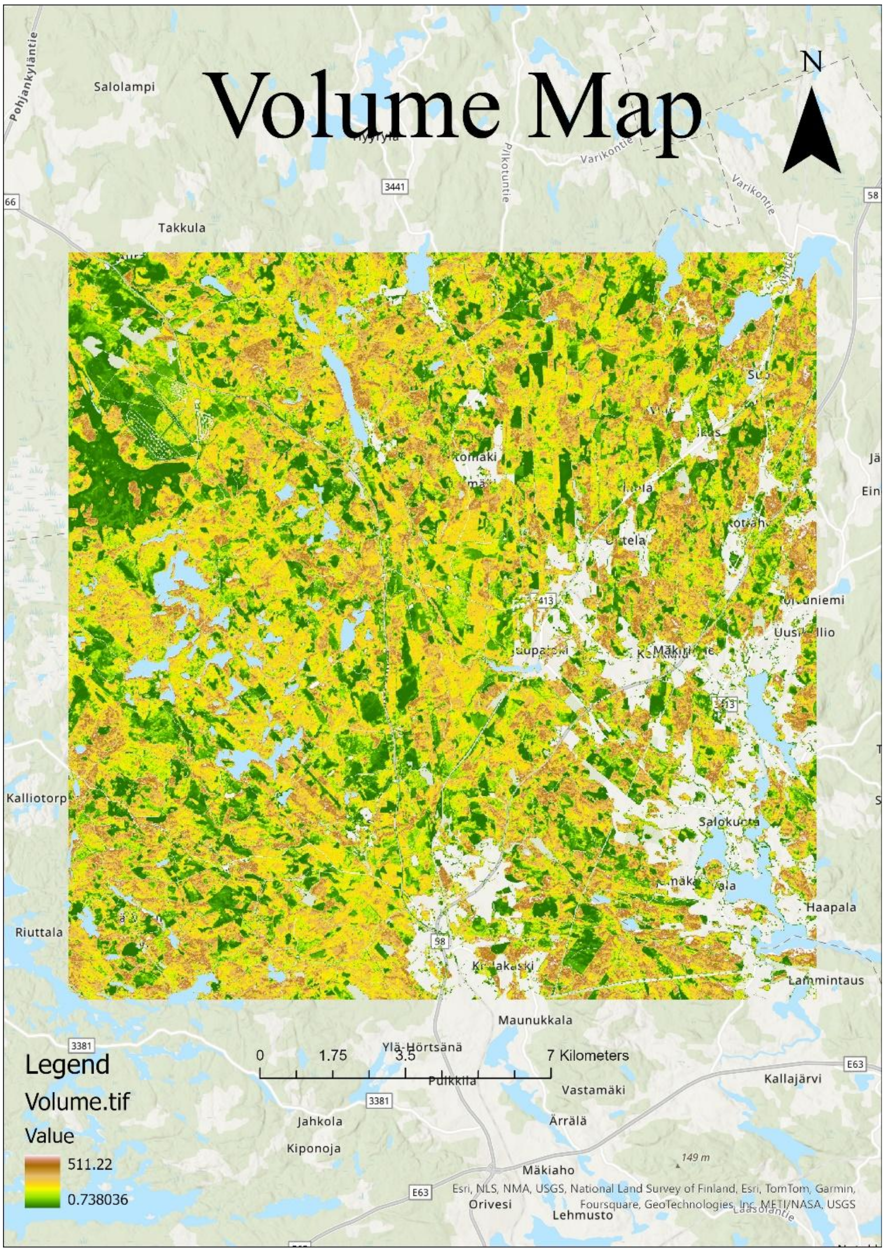
Volume Histograms and Interpretation

The volume histogram generated from the S2P2 model predictions provides a visual representation of the distribution of volume estimates across the Juupajoki area. It illustrates the frequency of occurrence of different volume ranges as predicted by the model. The distribution appears to be bimodal, with two prominent peaks, one near the lower end of the volume range and another just past the midpoint. This suggests that there are two prevalent volume classes within the dataset, which could correspond to different types of land cover or vegetation density. The right tail of the histogram extends towards larger volumes but with diminishing frequency, indicating that higher volume measurements are less common in the data. This bimodal distribution and the range of volume values provide insights into the variability within the dataset, which may impact the model's ability to predict volumes accurately across different classes.

Similarly, the volume histogram derived from the NFI map depicts the distribution of volume estimates across the same area, as recorded by the National Forest Inventory. The distribution is skewed to the right, with the highest frequency of volume measurements concentrated at the lower end of the scale. There appears to be a drop in frequency after the first bin, followed by a gradual rise that forms a peak before tapering off. This pattern suggests a commonality of lower volume measurements within the dataset and fewer occurrences of higher volumes, which may indicate a larger proportion of younger or less densely packed forests or other land cover types with lower biomass volumes. The long right tail highlights the presence of some areas with significantly higher volumes, though these are less frequent. The shape of the distribution is indicative of varied forest structures and possibly a mix of different forest age classes or types within the area covered by the NFI.

Consistency between the mean volumes and similar shapes of volume histograms may suggest a reasonable agreement between the S2P2 model predictions and the NFI map in estimating forest volumes for the Juupajoki area.

The predicted volume map



The predicted volume map displays spatial variations in volume across the landscape, as indicated by the colour-coded legend. The colors range from dark green (low volume) to light brown (high volume), with intermediate volumes shown in shades of yellow and orange. Areas with higher predicted volumes, representing mature or dense forests, are less widespread than areas with lower volumes, which dominate the landscape and may signify younger forests or other vegetative land cover. The spatial distribution pattern observed in the map can inform forest management and conservation efforts, highlighting regions with potentially high biomass and carbon storage capacity.

TASK 3

Predictor variables selected for the KNN model

In the KNN model, 'green', 'nir', 'nirnarrow', 'swir1', and 'swir2' are selected as the predictor variables for the model. These variables likely represent different spectral bands that are influential in predicting the volume of pine, spruce, and deciduous trees. The selection of these variables suggests that a combination of reflectance properties related to vegetation health and water content are significant predictors for tree volume estimation in this model.

- The 'green' band captures the reflection from vegetation
- The 'nir' represents the near-infrared spectrum which is typically absorbed by healthy vegetation
- The 'swir' bands might capture moisture content
- The inclusion of 'nirnarrow', a narrower band within the near-infrared spectrum, indicates a particular sensitivity to certain vegetative properties.

Species-specific RMSEs and scatterplots of observed vs. predicted values

```
> rmse(pred$pinevol.o,pred$pinevol)
[1] 74.04731
> relrmse(pred$pinevol.o,pred$pinevol)
[1] 1.138086
```

```
> rmse(pred$sprucevol.o,pred$sprucevol)
[1] 81.26861
> relrmse(pred$sprucevol.o,pred$sprucevol)
[1] 0.8831189
```

```
> rmse(pred$decidvol.o,pred$decidvol)
[1] 36.68648
> relrmse(pred$decidvol.o,pred$decidvol)
[1] 1.250574
```

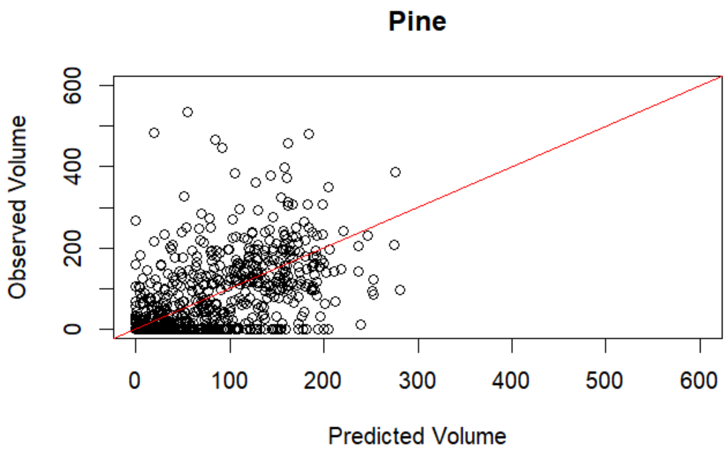


Figure 7: Scatter plot of predicted vs. observed Pine Volume

A relative RMSE of 113.8086% provides quantitative evidence for the large dispersion of data points in the pine volume predictions scatter plot, which indicates a significant difference between the actual and projected values. When the RMSE number is higher than 100%, it usually indicates that the average prediction error is higher than the mean observed volume and that the model's predictions are not very accurate. The scatter around the diagonal line, which represents the line of perfect agreement, indicates that either the dataset contains significant noise or complex patterns that are too complex for the KNN approach and the selected features to adequately model, or the KNN model may not be capturing the underlying patterns for pine volume estimation. Reevaluating the feature selection or model choice may be necessary due to the unreliability of the model's predictions for pine volume, as indicated by the high relative RMSE and scatter plot pattern.

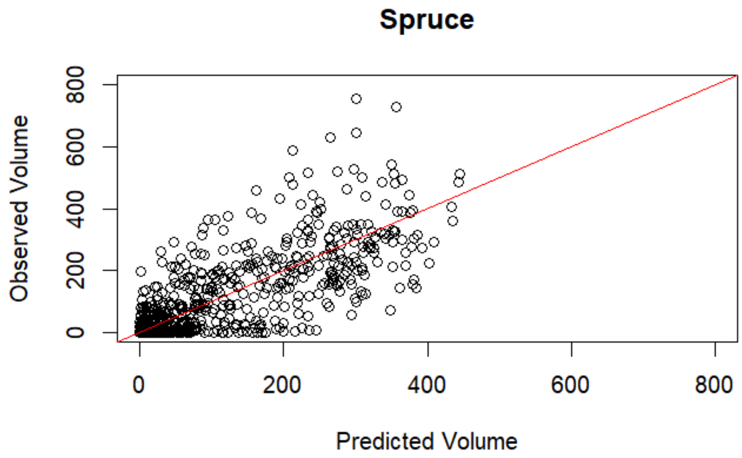


Figure 8: Scatter plot of predicted vs. observed Spruce Volume

In comparison to the preceding pine figure, the spruce volume estimation scatter plot displays forecasts compared against actual observations. The data points are slightly more tightly packed around the diagonal line, but they still show a noticeable spread. The spruce volume relative root mean square error (RMSE) is high (88.31189%), meaning that, on average, the projections differ significantly from the measured volumes—albeit less significantly than the pine volume estimation. While the dispersion grows with greater volume values, the concentration of dots closer to the origin shows the model is more accurate with smaller volume forecasts. This could mean that the model becomes less accurate at greater volume levels, or that the dataset's intrinsic higher spruce volume variability makes it harder for the KNN model to make correct predictions.

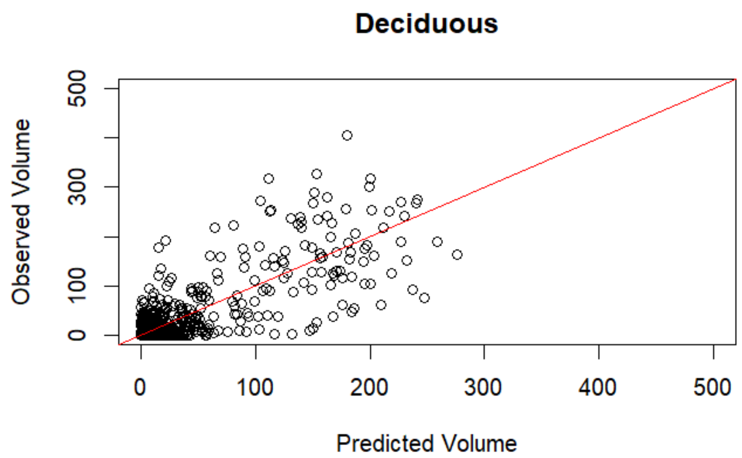
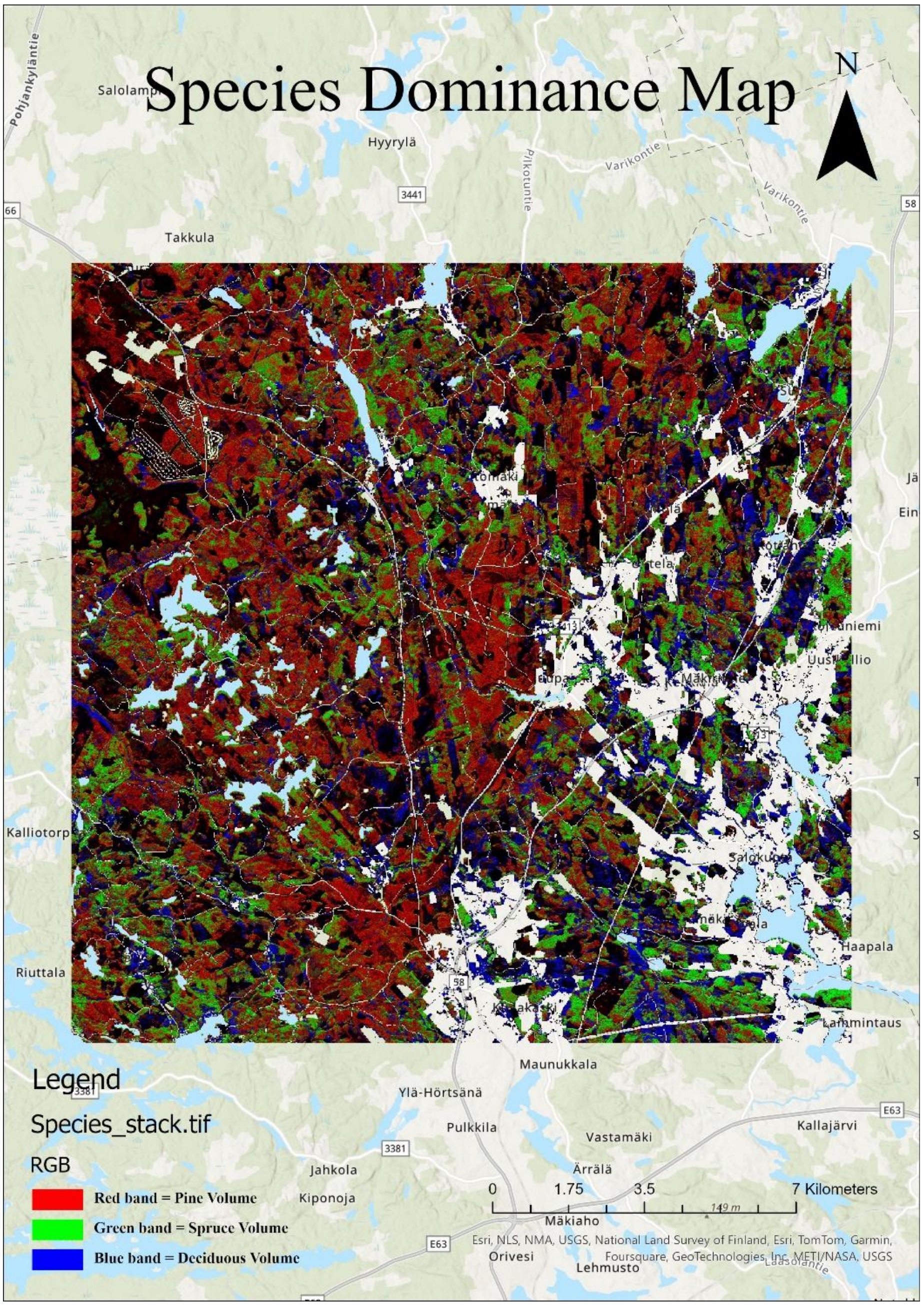


Figure 9: Scatter plot of predicted vs. observed Deciduous tree Volume

The deciduous volume estimation scatter plot and the corresponding relative RMSE of 125.0574% show that the model is not performing up to par, as evidenced by the significant departure from the line of perfect prediction. When the relative RMSE is greater than 100%, it indicates that the model is doing poorly because the average prediction errors are higher than the mean observed volume. The majority of data points are clustered towards the lower end of the scale, suggesting that either the dataset contains a large number of low-volume observations or the model may be underestimating larger volumes. The scatter of points denotes inconsistent predictions, suggesting that an alternative modeling strategy may be required or that the model was unable to adequately capture the intricacies or patterns of deciduous volume within the dataset.

Species dominance map for the Juupajoki area



The species dominance map represents three different bands that correspond to three different species volumes. The red band represents the pine volume, the green band represents the spruce volume, and the blue band represents the deciduous tree volume. From the raster volume map, it is quite evident that pine species has the highest volume followed by spruce and deciduous trees.